*Lab5.1.py - C:/Users/Vaishnavi PM/AppData/Local/Programs/Python/Python314/Lab5.1.py (3.14.2)*

File Edit Format Run Options Window Help

```python
import pandas as pd
data = pd.Series([10,20,30,40])
print(data)

import pandas as pd
data = pd.Series([10,20,30],index=["x","y","z"])
print(data)

import pandas as pd
data = {
    "Name": ["Ram", "Vani"],
    "Age": [20,25 ]
}
df = pd.DataFrame(data)
print(df)
print("\n")
#print(df.Name) #  or df["Name"]   # access based on col
print(df.loc[0]) # or df.loc[0,"col_Name"]  # 0th-row, Particular Col    #acess base on row
#print(df.loc[1])

import pandas as pd
df=pd.read_csv("C:\\Users\\Vaishnavi PM\\OneDrive\\Desktop\\Book3.csv")
print(df)

print(df.isnull())
print(df.dropna())
print(df.fillna(0))

# basic statistics method
import pandas as pd
print(df.MARKS.mean())
print(df.MARKS.sum())
print(df.MARKS.min())
print(df.MARKS.max())
print(df.MARKS.count())
```

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```python
import pandas as pd
data = {
    "Name": ["Alice", "Bob", "Charlie", "David", "Eva"],
    "Age": [24, 27, 22, 32, 29],
    "City": ["New York", "Los Angeles", "Chicago", "Houston", "Phoenix"]
}
df = pd.DataFrame(data)
print("Shape of df:", df.shape)        # (rows, columns)
print("Number of dimensions:", df.ndim)
print("Length of df:", len(df))        # Number of rows

print("\nFirst 3 rows:")
print(df.head(3))                      # First n rows

print("\nLast 2 rows:")
print(df.tail(2))                     # Last n rows

print("\nIndex of df:")
print(df.index)

print("\nColumns of df:")
print(df.columns)


import pandas as pd
data = {
"ID": [1, 2, 3, 4],
"Name": ["Rahul", "Priya", "Amit", "Sara"],
"Marks": [85, 90, 78, 92]}
#Convert to table (DataFrame)
df = pd.DataFrame(data)
#Save as CSV file
df.to_csv("students.csv",index=False)
print("CSV file created successfully")
```

Ln: 35 Col: 23

```
>>>
================ RESTART: C:/Users/Vaishnavi PM/AppData/Local/Programs/Python/Python314/Lab5.1.py ================
0    10
1    20
2    30
3    40
dtype: int64
x    10
y    20
z    30
dtype: int64
   Name  Age
0   Ram   20
1  Vani   25


Name    Ram
Age      20
Name: 0, dtype: object
   Unnamed: 0   NAME CLASS  MARKS
0           0  Vaish   BCA   89.0
1           1  bujji   BCA   95.0
2           2    mru   BCA   83.0
3           3  ruthu   BCA    NaN
   Unnamed: 0   NAME  CLASS  MARKS
0       False  False  False  False
1       False  False  False  False
2       False  False  False  False
3       False  False  False   True
   Unnamed: 0   NAME CLASS  MARKS
0           0  Vaish   BCA   89.0
1           1  bujji   BCA   95.0
2           2    mru   BCA   83.0
   Unnamed: 0   NAME CLASS  MARKS
0           0  Vaish   BCA   89.0
```

IDLE Shell 3.14.2

File Edit Shell Debug Options Window Help

```
   Unnamed: 0   NAME CLASS  MARKS
0           0  Vaish   BCA   89.0
1           1  bujji   BCA   95.0
2           2    mru   BCA   83.0
   Unnamed: 0   NAME CLASS  MARKS
0           0  Vaish   BCA   89.0
1           1  bujji   BCA   95.0
2           2    mru   BCA   83.0
3           3  ruthu   BCA    0.0
89.0
267.0
83.0
95.0
3
Shape of df: (5, 3)
Number of dimensions: 2
Length of df: 5

First 3 rows:
      Name  Age         City
0    Alice   24     New York
1      Bob   27  Los Angeles
2  Charlie   22      Chicago

Last 2 rows:
    Name  Age     City
3  David   32  Houston
4    Eva   29  Phoenix

Index of df:
RangeIndex(start=0, stop=5, step=1)

Columns of df:
Index(['Name', 'Age', 'City'], dtype='str')
CSV file created successfully
>>>
```